

Computer Networks

Credit Hours: 3-0

Course Outline

Introduction to Computer Networks, Network Models and Topologies, The OSI and TCP/IP Reference Models, Data Transmission Concepts and Techniques, Transmission Media, Data Link Layer and Protocols, Error Detection and Correction, Flow and Error Control Protocols, Medium Access Control (MAC) Techniques, Local Area Networks (LANs) and Ethernet, Network Layer and IP Addressing, Routing Algorithms and Protocols, Transport Layer Protocols (TCP and UDP), Congestion Control, Application Layer Protocols (HTTP, DNS, FTP, SMTP), Network Security Fundamentals, Wireless and Mobile Networks, Emerging Networking Technologies (IoT, Cloud Networking, 5G)

16-Week Course Plan

Week 1

****Week 1: Introduction to Computer Networks****

****Distribution:** 3-0**

Lecture 1: Introduction to Computer Networks

Learning Objectives

- * Define computer networks and their importance
- * Identify the key concepts and terminology in computer networking

Concepts

Computer networks are collections of interconnected devices that communicate with each other to share resources and exchange data. The importance of computer networks lies in their ability to enable communication, collaboration, and data sharing between devices. Key concepts and terminology in computer networking include:

- * Devices: computers, servers, printers, routers, switches, etc.
- * Networks: local area networks (LANs), wide area networks (WANs), metropolitan area networks (MANs), etc.
- * Protocols: rules and standards that govern data transmission and communication
- * Topologies: physical and logical arrangements of devices in a network

Lecture 2: Network Models and Topologies

Learning Objectives

- * Describe the OSI and TCP/IP reference models
- * Identify common network topologies

Concepts

The OSI (Open Systems Interconnection) reference model is a 7-layered framework that describes how data is transmitted over a network. The TCP/IP (Transmission Control Protocol/Internet Protocol) reference model is a 4-layered framework that is widely used in the internet.

Common network topologies include:

- * Bus topology: a single cable connects all devices
- * Star topology: a central device connects to all other devices
- * Ring topology: devices are connected in a circular configuration
- * Mesh topology: each device connects to every other device

Lecture 3: Introduction to Network Fundamentals

Learning Objectives

- * Describe the importance of data transmission and reception
- * Identify the key concepts in data transmission concepts and techniques

Concepts

Data transmission and reception are essential for computer networks to function. Key concepts in data transmission concepts and techniques include:

- * Analog and digital signals
- * Data encoding and decoding
- * Data transmission techniques: modulation, multiplexing, etc.
- * Error detection and correction techniques: checksums, CRCs, etc.

Lab: None

Week 2

Week 2: Network Models and Topologies (3-0)

Lecture 1: Introduction to Network Models and Topologies

Network models and topologies are essential concepts in understanding computer networks. A network model represents the structure and organization of a network, while a topology refers to the physical or logical arrangement of devices and connections.

Learning Objectives:

- Define network models and topologies.
- Identify different network models (e.g., OSI, TCP/IP).
- Describe common network topologies (e.g., bus, star, ring).

Concepts:

- Network models: OSI, TCP/IP, and others.
- Network topologies: bus, star, ring, mesh, and hybrid.

Lecture 2: Network Topologies

Network topologies determine how devices are connected and communicate with each other. Common topologies include bus, star, ring, and mesh.

Learning Objectives:

- Compare and contrast different network topologies.
- Identify advantages and disadvantages of each topology.
- Describe how topologies affect network performance.

Concepts:

- Bus topology: a single cable connects all devices.
- Star topology: a central device connects to all others.
- Ring topology: devices connect in a circular configuration.
- Mesh topology: each device connects directly to every other device.

Lecture 3: Network Models and Topologies in Practice

Network models and topologies are used in various networking scenarios, including local area networks (LANs) and wide area networks (WANs).

Learning Objectives:

- Apply network models and topologies to real-world scenarios.
- Identify the importance of network models and topologies in network design.
- Describe how network models and topologies affect network performance.

Concepts:

- LANs: use bus, star, or ring topologies.
- WANs: use point-to-point or mesh topologies.
- Network design: consider network models and topologies when designing a network.

Week 3

****Week 3: Network Layer and IP Addressing****

****3-0****

Lecture Title

Network Layer and IP Addressing

Learning Objectives

- * Understand the role of the Network Layer in the OSI model
- * Learn about IP addressing and its significance
- * Understand the concept of subnetting and supernetting

Concepts

The Network Layer is the third layer of the OSI model, responsible for routing data between devices on different networks. It provides logical addressing and routing functions. The most widely used protocol at this layer is the Internet Protocol (IP).

IP Addressing is a crucial concept in networking. An IP address is a unique identifier assigned to each device on a network. It consists of a network ID and a host ID. The network ID identifies the network, while the host ID identifies the device on that network.

Subnetting is a technique used to divide a large network into smaller, more manageable networks called subnets. This is achieved by borrowing bits from the host ID to create a subnet mask. Supernetting is the opposite of subnetting, where a large number of subnets are combined into a single network.

The Network Layer also handles error-free transfer of data between devices, but it does not guarantee delivery. This is the responsibility of the Transport Layer.

Week 4

Week 4: Network Layer and IP Addressing (2-0)

Lecture Title: Network Layer and IP Addressing

Learning Objectives

- Understand the Network Layer's role in routing and packet forwarding
- Learn about IP addressing and subnetting
- Understand the different types of IP addresses

Concepts

The Network Layer is responsible for routing and packet forwarding, using logical addresses to forward packets between devices. The Internet Protocol (IP) is the primary protocol used in the Network Layer, providing logical addressing and routing functionality.

IP Addressing

- **IP Address**: A 32-bit or 128-bit number used to uniquely identify a device on a network
- **IPv4**: 32-bit address, divided into 4 octets (e.g., 192.168.1.1)
- **IPv6**: 128-bit address, divided into 8 groups of 16 bits (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
- **Subnetting**: Dividing an IP address into a network and host parts to create subnetworks

Subnet Mask

- A 32-bit number used to identify the number of bits used for the network and host parts of an IP address
- Used to determine which IP addresses are on the same subnet

Default Gateway

- The device that connects two or more networks and forwards packets between them
- Typically the router in a network

Summary

The Network Layer uses logical addresses to forward packets between devices, and IP addresses provide unique identification for each device. Subnetting and subnet masks are used to create subnetworks, and the default gateway connects networks and forwards packets.

Week 5

****Week 5: Network Layer and IP Addressing (3-0)****

Lecture Title: Network Layer and IP Addressing

Learning Objectives

- * Understand the Network Layer's role in routing packets
- * Learn about IP Addressing and its importance
- * Familiarize with IPv4 and IPv6 address formats

Concepts

The Network Layer is responsible for routing packets between devices on a network. It provides logical addressing, which enables devices to communicate with each other. IP Addressing is a critical aspect of the Network Layer, as it allows devices to be uniquely identified on a network.

IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses. IPv4 addresses are typically represented in dotted decimal notation (e.g., 192.168.1.1), while IPv6 addresses use hexadecimal notation (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

Key concepts include:

- * IP Address classes (A, B, C)
- * Subnetting and supernetting
- * DHCP and static IP addressing
- * IPv6 features and benefits

Understanding the Network Layer and IP Addressing is crucial for building and managing computer networks.

Week 6

Lecture Title: Network Layer and IP Addressing

Learning Objectives

- Understand the Network Layer's role in the OSI model
- Describe the functions of the Network Layer
- Explain the concept of IP addressing and IP addressing schemes
- Identify the different types of IP addresses (IPv4 and IPv6)

Concepts

The Network Layer is the third layer of the OSI model, responsible for routing data between different networks. It provides logical addressing and routing functions that enable data to be delivered between devices on different networks. The Network Layer uses logical addresses called IP addresses to identify devices on a network.

IP addressing is a method of assigning a unique address to each device on a network. The most common types of IP addresses are IPv4 and IPv6. IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses. IPv6 is designed to provide a larger address space and improved security features.

There are two types of IP addresses:

- Unicast addresses: Used to send data to a single device
- Multicast addresses: Used to send data to a group of devices
- Broadcast addresses: Used to send data to all devices on a network

Understanding IP addressing is crucial for network configuration and troubleshooting.

Week 7

****Week 7: Network Layer and IP Addressing****

Lecture Title: Network Layer and IP Addressing

Learning Objectives:

- Understand the Network Layer's role in packet forwarding.
- Learn about IP addressing and its significance.
- Recognize the types of IP addresses and their uses.

Concepts:

The Network Layer is the third layer of the OSI model, responsible for packet forwarding between devices. It provides logical addressing, routing, and congestion control. The most widely used protocol at this layer is the Internet Protocol (IP).

IP addressing is a way to uniquely identify devices on a network. IP addresses are 32-bit or 64-bit numbers assigned to devices, allowing them to communicate with each other. There are two types of IP addresses:

- ****IPv4****: 32-bit addresses, typically written in dotted decimal notation (e.g., 192.168.1.1).
- ****IPv6****: 64-bit addresses, written in hexadecimal notation (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).

IP addresses are divided into two main categories:

- ****Public IP addresses****: Used for devices connected to the internet, ensuring unique addresses for online communication.
- ****Private IP addresses****: Used for devices within a private network, allowing for internal communication without direct internet access.

Understanding IP addressing is crucial for network communication and configuration.

Week 8

Week 8: Network Layer and IP Addressing (3-0)

Lecture 1: Network Layer Overview (1.5 hours)

Network Layer Overview

Learning Objectives

- Understand the primary functions of the network layer
- Identify the key components of the network layer

Concepts

The network layer is responsible for routing data between devices on different networks. It provides the logical addressing of data packets and decides the path they should take to reach their destination. The key components of the network layer are:

- Routing: The process of forwarding data packets between networks
- Logical Addressing: Assigning a unique address to each device on the network
- Path Determination: Determining the best path for data to travel from source to destination

The network layer uses routing protocols to exchange routing information between devices and make routing decisions.

Lecture 2: IP Addressing (1.5 hours)

IP Addressing

Learning Objectives

- Understand the basics of IP addressing
- Identify the different types of IP addresses
- Explain the concept of subnetting

Concepts

IP addressing is a method of assigning a unique address to each device on a network. There are two main types of IP addresses:

- IPv4: Uses 32-bit addresses, divided into four octets separated by dots (e.g., 192.168.1.1)
- IPv6: Uses 128-bit addresses, written in hexadecimal format (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)

Subnetting is the process of dividing a network into smaller sub-networks by creating sub-network masks.

Lecture 3: Routing Algorithms and Protocols (1 hour)

Routing Algorithms and Protocols

Learning Objectives

- Understand the basics of routing algorithms and protocols

- Identify the different types of routing algorithms and protocols
- Explain the concept of distance-vector routing

Concepts

Routing algorithms and protocols are used to exchange routing information between devices and make routing decisions. There are two main types of routing algorithms:

- Distance-Vector Routing: Uses the distance to a network to determine the best path (e.g., RIP, IGRP)
- Link-State Routing: Uses information about the network topology to determine the best path (e.g., OSPF, EIGRP)

Routing protocols are used to exchange routing information between devices, including RIP, OSPF, EIGRP, and BGP.

Week 9

****Week 9: Network Layer and IP Addressing (3-0)****

Lecture Title: Network Layer and IP Addressing

Learning Objectives

- * Understand the functions of the Network Layer
- * Learn about IP Addressing concepts and protocols
- * Identify the types of IP addresses and their uses

Concepts

The Network Layer is the third layer of the OSI Reference Model and is responsible for routing data between different networks. It provides connectivity between devices on different networks and is the layer at which routing occurs.

IP Addressing is a crucial concept in the Network Layer. IP addresses are unique addresses assigned to devices on a network, allowing them to communicate with each other. There are two main types of IP addresses: IPv4 and IPv6. IPv4 addresses are 32-bit addresses, while IPv6 addresses are 128-bit addresses.

IPv4 addresses are typically written in dotted decimal notation, with four numbers separated by dots. IPv6 addresses are written in hexadecimal notation, with eight groups of four hexadecimal digits separated by colons. IP addresses can be classified into two categories: public and private. Public IP addresses are unique and can be accessed from the internet, while private IP addresses are used within a network and are not accessible from the internet.

IP Addressing protocols include DHCP (Dynamic Host Configuration Protocol) and DNS (Domain Name System). DHCP assigns IP addresses to devices on a network, while DNS translates domain names into IP addresses.

****Key Terms:**** Network Layer, IP Addressing, IPv4, IPv6, Public IP, Private IP, DHCP, DNS.

Week 10

****Week 10: Network Layer and IP Addressing (3-0)****

Lecture Title: Network Layer and IP Addressing

Learning Objectives:

- Understand the Network Layer's role in routing data.
- Learn about IP addressing and its significance.

Concepts:

The Network Layer is responsible for routing data between devices on different networks. It ensures that data reaches its destination by breaking it into packets and assigning a logical address, known as an IP address. IP addresses are 32-bit (IPv4) or 128-bit (IPv6) numbers that uniquely identify devices on a network.

There are two main types of IP addresses: public and private. Public IP addresses are assigned by Internet Service Providers (ISPs) and are unique to each device on the internet. Private IP addresses, on the other hand, are used within private networks and are not routable on the internet.

The Network Layer also uses routing tables to determine the best path for data to travel between networks. Routing protocols, such as RIP and OSPF, are used to maintain these tables and ensure efficient data transfer.

Key Terms:

- Network Layer
- IP Addressing
- Public IP Addresses
- Private IP Addresses
- Routing Tables
- Routing Protocols

Week 11

****Week 11: Transport Layer Protocols (TCP and UDP)****

Lecture Title: Transport Layer Protocols

Learning Objectives:

- Understand the Transport Layer's role in network communication
- Compare and contrast TCP and UDP protocols
- Identify the key features of TCP and UDP

Concepts:

The Transport Layer is responsible for reliable data transfer between devices. Two primary protocols operate at this layer: TCP (Transmission Control Protocol) and UDP (User Datagram

Protocol).

TCP is a connection-oriented protocol ensuring reliable data transfer. It establishes a connection between devices before data transfer and guarantees delivery of data packets in the correct order. TCP uses sequence numbers and acknowledgments to ensure error-free communication.

UDP is a connectionless protocol, providing best-effort delivery. It does not establish a connection and does not guarantee delivery or order of packets. UDP is often used for real-time applications, such as video streaming, where speed is more critical than reliability.

Key features of TCP include:

- * Connection establishment and termination
- * Reliable data transfer
- * Flow control
- * Error detection and correction

Key features of UDP include:

- * Connectionless communication
- * Best-effort delivery
- * No guarantee of delivery or order of packets
- * Faster data transfer due to lack of connection establishment and termination overhead

Week 12

Week 12: Network Security Fundamentals

Lecture Title: Network Security Fundamentals

Learning Objectives

- Understand the importance of network security
- Identify types of network security threats
- Learn about common network security measures

Concepts

Network security is a critical aspect of computer networking that protects against unauthorized access, use, disclosure, disruption, modification, or destruction of network resources. Network security threats include:

- * Unauthorized access
- * Data breaches
- * Malware and viruses
- * Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks
- * Man-in-the-Middle (MitM) attacks

Common network security measures include:

- * Firewalls
- * Encryption
- * Authentication and authorization
- * Intrusion Detection and Prevention Systems (IDPS)
- * Secure communication protocols (HTTPS, SSH, SFTP)

Understanding network security fundamentals is essential for designing and implementing secure network architectures.

Week 13

****Week 13: Network Security Fundamentals (3-0)****

Lecture Title: Network Security Fundamentals

Learning Objectives:

- Understand network security threats and vulnerabilities
- Identify common network security protocols and mechanisms
- Analyze the importance of network security in computer networks

Concepts:

Network security is a critical aspect of computer networks. With the increasing reliance on networks for communication, data transfer, and business operations, the risk of security breaches and attacks has grown significantly. Network security threats can be categorized into two main types: unauthorized access and data tampering.

Unauthorized access includes attacks such as hacking, phishing, and malware. Data tampering involves modifying or deleting data without authorization. Common network security protocols and mechanisms include firewalls, intrusion detection and prevention systems, encryption, and access control lists.

Firewalls act as a barrier between the internal network and the external network, blocking unauthorized access and malicious traffic. Intrusion detection and prevention systems monitor network traffic for signs of unauthorized access and take action to prevent it. Encryption ensures that data transmitted over the network is secure and cannot be intercepted or read by unauthorized parties.

Access control lists (ACLs) regulate who can access the network and what resources they can access. Network security is essential in preventing data breaches, protecting sensitive information, and maintaining the integrity of the network.

****Reading Assignment:**** Chapters 12-13 in the textbook.

****Quiz:**** (Will be available on the course learning management system)

Week 14

Week 14: Transport Layer Protocols (TCP and UDP)

3-0

Lecture 1: Introduction to TCP/IP Transport Layer Protocols

TCP and UDP Overview

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are two main transport-layer protocols used in the internet. TCP provides reliable, connection-oriented, and error-checked data transfer between devices, whereas UDP provides best-effort, connectionless, and unreliable data transfer.

Learning Objectives:

- Explain the difference between TCP and UDP.
- Describe the characteristics of TCP and UDP.

Concepts:

- TCP vs. UDP
- TCP characteristics: connection-oriented, reliable, and error-checked
- UDP characteristics: connectionless, best-effort, and unreliable

Lecture 2: TCP Connection Establishment, Data Transfer, and Termination

TCP Connection Establishment

TCP connection establishment involves three-way handshake between the client and server. The process includes:

- SYN (synchronize) packet from the client to the server
- SYN-ACK (synchronize-acknowledgment) packet from the server to the client
- ACK (acknowledgment) packet from the client to the server

Learning Objectives:

- Explain the steps involved in TCP connection establishment.
- Describe the significance of TCP connection establishment.

Concepts:

- TCP three-way handshake
- SYN, SYN-ACK, and ACK packets
- Connection establishment significance

Lecture 3: TCP Connection Termination

TCP Connection Termination

TCP connection termination involves four-way handshake between the client and server. The process includes:

- FIN (finish) packet from the client to the server
- ACK (acknowledgment) packet from the server to the client
- FIN (finish) packet from the server to the client
- ACK (acknowledgment) packet from the client to the server

Learning Objectives:

- Explain the steps involved in TCP connection termination.
- Describe the significance of TCP connection termination.

Concepts:

- TCP four-way handshake
- FIN and ACK packets
- Connection termination significance

Week 15

****Week 15: Emerging Networking Technologies (IoT, Cloud Networking, 5G)****

Lecture Title: Emerging Networking Technologies

Learning Objectives

- * Understand the concepts of Internet of Things (IoT) and its applications
- * Learn about Cloud Networking and its benefits
- * Familiarize with the basics of 5G wireless networks

Concepts

The Internet of Things (IoT) refers to the network of physical devices, vehicles, home appliances, and other items embedded with sensors, software, and connectivity, allowing them to collect and exchange data with other devices and systems over the internet. IoT has various applications, including smart homes, cities, and industries.

Cloud Networking involves the use of cloud computing resources to provide network services, such as storage, processing, and security. This approach enables scalable, on-demand network resources and improved network management.

5G wireless networks are the latest generation of cellular networks, offering faster data speeds, lower latency, and greater connectivity than previous generations. 5G enables applications such as mission-critical communications, massive machine-type communications, and ultra-high-definition video streaming.

These emerging networking technologies have the potential to transform various industries and aspects of our lives, and it is essential to understand their concepts and applications.

****No lab this week****

Week 16

****Week 16: Emerging Networking Technologies****

Lecture Title: Emerging Networking Technologies

Learning Objectives:

- Understand the concept of IoT and its applications
- Learn about cloud networking and its benefits
- Familiarize with 5G wireless network technology

Concepts:

Emerging networking technologies are revolutionizing the way we communicate and interact with each other. These technologies have the potential to transform various industries and aspects of our lives.

Internet of Things (IoT)

IoT refers to the network of physical devices, vehicles, home appliances, and other items embedded with sensors, software, and connectivity, allowing them to collect and exchange data. IoT has numerous applications, including smart homes, industrial automation, and healthcare.

Cloud Networking

Cloud networking involves the use of cloud computing and storage to provide scalable and on-demand network resources. Cloud networking offers numerous benefits, including reduced capital expenditures, increased flexibility, and improved scalability.

5G Wireless Network

5G is the fifth generation of wireless network technology, offering faster data rates, lower latency, and greater connectivity than its predecessors. 5G will enable a wide range of applications, including enhanced mobile broadband, mission-critical communications, and massive machine-type communications.

****Key Takeaways:****

- * IoT, cloud networking, and 5G are emerging networking technologies that will shape the future of communication and interaction.
- * IoT has numerous applications across various industries.
- * Cloud networking offers benefits such as reduced capital expenditures and increased flexibility.
- * 5G will enable faster data rates, lower latency, and greater connectivity than its predecessors.

****References:****

- * [List relevant sources, e.g., books, articles, websites]